EX.NO:-12(b) chat client-Server using TCP/IP sockets

AIM:-

To write and execute a C/java socket program to implement a two way chat client and chat server and the Server correctly echoed the received message back to the client.

ALGORITHM:-

1. Create a Socket bind it to a designed port.
2. Listen and accept connection from the client.
3. Start the communication loop.
4. Close the saket when chat ends.

Client Slide :-

1- Create a socket and connect to the server Ip and port.
2. start a communication loop.
3. Close the socket connection.

TCP Server

```
import Socket
S = Socket.Socket()
S = bind(('localhost', 5000))
S.listen(1)
c, addr = S.accept()
while True:
    msg = c.recv(1024).decode()
    if msg == "bye":
        break
    print("client :", msg)
    reply = input("Server")
    c.send(reply.encode())
c.close()
S.close()
```

TCP client

```
import socket
s=socket.socket()
s connect ((''localhost':5000))
while True:
    msg = input("client:")
    s.send(msg.encode())
        break
    reply.s recv(1024).decode()
s.close()
```

```
import socket
S=socket.Socket(SocketAF.INET, socket sock.DGRAM)
S = bind (('local host = 5000))
while true
    msg.add = s.recv from (1024)
    msg = msg = decode()
    print ("client:", msg)
        if msg = "bye"
        break
S.close()
```

```
import socket
S=socket.socket(Socket AF_INET, socket, DGRAM)
while True
    msg = input ("client:")
S.sendto (msg.encode(), (('localhost:5000))
if msg = "bye"
    break
S.close()
```

OUTPUT:-

TCP chat server

TCP chat server started

client connected

client : Hello

client : bye

chat ended

Tcp chat client

client : Hello

Server : Hi

client : bye

chat ended

UDP chat server

UDP chat Server started

waiting for Messages

client : Hello

Server : Hi

client : bye

chat ended

UDP chat client

client : Hello

Server : Hi

client : bye

chat ended

RESULT :-

Chat communication between the client and server was successfully implemented using .TCP and UDP Socket in python